

Reduction of estimation error impact in the risk parity strategies

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We consider the risk parity strategy in the presence of estimation errors. We show that risk contributions from constituents of this portfolio can be considerably sensitive to estimation errors in the sense that risk contributions are highly uneven on an ex post basis. In particular, we demonstrate that the sensitivity becomes exaggerated if Fama-French factors constitute the portfolio because of their characteristic of having low pairwise correlations. Our work demonstrates that the instability of the out-of-sample risk contributions is associated with a local property with statistical significance near to the constructed portfolio. Based on this observation, we propose a new algorithm for the risk parity strategy to mitigate the sensitivity of the optimized portfolio's out-of-sample risk contributions from estimation errors. Through empirical study, we find that the portfolio constructed by the proposed algorithm consistently outperforms its competitors in terms of the out-of-sample risk contributions.

Keywords: Equal risk contribution; Risk parity; Estimation error; Estimation error sensitivity; Resampling method

JEL Classification: C61, G11

1. Introduction

One of the most commonly encountered problematic issues in modern portfolio optimization is the estimation errors in the moments of returns. With the presence of estimation errors, the optimized portfolio is hardly efficient out of sample (Jobson and Korkie 1981, Michaud 1989). In particular, it is known that means of the returns are more difficult to estimate than covariances (Merton 1980) and small changes in the means can lead to large fluctuations in the optimized portfolio's weights (Best and Grauer 1991, Chopra and Ziemba 1993). Due to the difficulty in estimating the means of returns and the associated estimation error's largely unpredictable effect on the optimized portfolio's out-of-sample efficiency, investors question the value of incorporating the estimated expected returns into their portfolio optimization procedure to target a certain level of return in the sense of the classical mean-variance optimization framework (Markowitz 1952).

Moreover, during the 2007–2008 global financial crisis, investors witnessed the failure of their portfolios that were believed to be diversified. This led them to cast doubt on classical approaches to risk diversification, implemented via, for

example, 60/40 allocation to stocks and government bonds or mean-variance optimization imposed with additional heuristic constraints. The recent global market meltdown caused by the COVID-19 pandemic made this question again pressing. Specifically, not only did we observe that the traditional asset classes, e.g. stocks and bonds, sharply fell in tandem, but also that bonds were highly volatile which substantiated that mere exposure to bonds cannot hedge equity risk (Ren 2020). Despite some criticism of risk diversification in itself, however, most people in the investment community believe that the attacks are undeserved and risk diversification is still an important investment principle to pursue (Lee 2011, Ilmanen and Kizer 2012).

In this regard, there has been considerable effort for devising better approaches to portfolio construction where the primary interest focuses on risk management. These approaches, collectively labeled as risk-based asset allocation, do not require forecasts of expected returns nor seek a certain level of portfolio return explicitly. Rather, it is risk diversification that is primarily emphasized. Risk parity (RP) strategy, also referred to as equal risk contribution (ERC) or risk budgeting (RB) (Qian 2005, Maillard *et al.* 2010, Roncalli 2013), is one of the main approaches stemmed from such efforts attracting the investment community over the last decade (Lee 2014). The only other portfolio that does not require expected return

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forecasts and is defined in the mean-variance optimization framework is the global minimum variance portfolio (Xing *et al.* 2014). Here, the ‘marginalized,’ and not the ‘total’ as in the RP portfolio, risk contribution by each asset in the portfolio is designed to be equal.

In the RP approach, a portfolio is constructed so that each asset in the portfolio contributes equal risk to the overall risk of the portfolio. A commonly used concept to express this is *risk contributions* of which the portfolio risk is decomposed as a sum. The concept of risk contribution is financially meaningful and mathematically well-defined (Tasche 2008), with which an optimization problem to find the corresponding portfolio can be defined. The RP portfolio has favorable properties in that the optimized capital allocation are well-diversified across the assets and remain restrictively affected by the change of asset covariances resulting from forecast errors. More importantly, it can achieve a higher risk-adjusted return by systematically reducing the concentration of risk. Indeed, the RP portfolio is mean-variance efficient if return and risk profiles satisfy some conditions, specifically, if all assets have the same Sharpe ratio and the correlation between any pair of assets is the same (Maillard *et al.* 2010). Even if the conditions are not strictly satisfied, using the RP strategy can be beneficial since it has been widely documented that Sharpe ratios are likely to be comparable across assets, over the long term (Lee 2014).

In practice, on the other hand, it may not be enough to merely achieve high Sharpe ratios. For example, it is often the case that an investor is required to achieve a certain level of returns (Lee 2014). For this reason, we consider Fama-French factors (Fama and French 1993, 1996, 2015) as our portfolio constituents in this paper. These factors are known to provide high returns in the long run, as evidenced in various places in the literature (Asness *et al.* 2013, Blitz and Vidojevic 2019, Dimson *et al.* 2017). Furthermore, they possess the favorable characteristic of having low correlation (Ilmanen and Kizer 2012) which affords them risk diversification that establishes them as good candidates for RP portfolio constituents.

The aim of this study is twofold. First, we clarify the impact of misspecification in the covariance estimates on the performance of RP portfolios in terms of the defining objective of the RP approach. Specifically, we investigate whether the out-of-sample risk contributions of a RP portfolio’s constituents are indeed equal. To this end, we evaluate the extent to which the a portfolio’s risk is diversified into its constituents by using an often used concentration measure. We demonstrate that the conventional RP portfolios composed of Fama-French factors may have a highly unreliable characteristic in terms of the out-of-sample risk contributions of portfolio constituents and verify why this happens by investigating the landscape of the concentration measure.

Second, we propose a remedy for enhancing the reliability of RP portfolios. Through the analysis of the landscape mentioned above, we show that there is useful information that we can exploit to make the actual risk contributions of constituents close to the true ones. Furthermore, we present that the information can be estimated using historical data in a credible manner. Using this observations, we propose an algorithm for RP portfolio construction that is inspired by

the resampling technique of (Michaud and Michaud 2008). We evaluate the validity of our proposed method by showing that our method consistently outperforms the conventional RP approach as well as other approaches aimed at mitigating the impact of the misspecification in covariance estimates.

The rest of the paper is organized as follows. In Section 2, we briefly review the sensitivity problem in portfolio optimization, focusing on the RP approach. In Section 3, we describe our problem of interest with illustrative empirical observations. We then explain our algorithm and its rationale, which is verified with a regression analysis. In Section 4, we evaluate the effectiveness of our method by comparing it with other benchmarks using various performance metrics including the measure of diversification of risk contributions from portfolio constituents. Finally, we conclude the paper in Section 5.

2. Related works

The first financial interpretation of the RP approach was given by (Qian 2005, 2006) who showed that, when a portfolio suffers from a large loss, the loss attributed to each constituent of the portfolio can be approximated by the risk contribution of the constituent. This observation suggests that the RP approach can be an effective way to prevent the overall portfolio risk from being dominated by a handful of portfolio constituents at the moment when risk diversification is truly needed. The work of (Maillard *et al.* 2010) is another milestone in theoretical study of the RP strategy, where it was proved that volatility of an RP portfolio is placed between those of two typical portfolios that are constructed irrespectively of asset returns, namely equally weighted and minimum variance portfolios. Kaya and Lee (2012) presented a set of optimality conditions of RP portfolios in the sense of mean-variance optimization and proposed a generalized utility function whose special case is maximized by an RP portfolio, through which they shed light on the investment rationale of the RP approach. Fisher *et al.* (2015) presented the minimax optimality in terms of returns and Sharpe ratios of RP portfolios by adopting notions of game theory in a simplified setting called naive risk parity strategy. Recent work of Ardia *et al.* (2018) extended the RP strategy by defining a novel performance metric for seeking a balance between performance contribution and risk contribution of each portfolio constituent. Based on this metric, they tried to tilt various risk-based portfolios, including RP, into the direction for which the performance and risk contributions are better aligned.

Our paper is closely related to the literature on portfolio sensitivity. The sensitivity of optimized portfolios to estimation errors in the classical mean-variance sense was extensively studied in the literature including (Jobson and Korkie 1981, Michaud 1989), to name a few. The work of Simaan and Simaan (2019) analytically investigated investors’ portfolio choice in relation to the level of estimation risks. They rationalized the fact that the portfolio optimized when the investors have full information about means and covariances of asset returns should be different when the level of estimation risks are high, e.g. when the investment

universe contains assets of emerging markets. Recently, the sensitivity problem in the RP approach has gained attention. Fisher *et al.* (2015) studied the sensitivity of RP portfolios using historical returns of two asset classes: stocks and bonds. Specifically, they calculated the vector angle between the weights of out-of-sample tangency portfolios and those of RP and in-sample tangency portfolios and showed that the angle between RP and out-of-sample tangency portfolios is likely to be lower than that between in-sample and out-of-sample tangency portfolios. Ardia *et al.* (2017) compared various risk-based strategies, including RP, by computing distance between portfolio weights constructed using true and estimated covariances while varying the amount of misspecification in the estimates via Monte Carlo simulations. Through the simulation study, they showed that RP portfolios are more robust than other risk-based approaches to the estimation errors in the covariance matrix unless the risk profiles are characterized by low average level of correlations between assets and a high level of dispersion between the correlation coefficients, where the risk contributions of the assets are far from diversified.

In addition to studying the impact of estimation errors, remedies to mitigate the impact have been proposed in the classical mean-variance setting under various methodologies, which include imposing constraints to the optimization problem (Jagannathan and Ma 2003, DeMiguel *et al.* 2009a, Xing *et al.* 2014), devising better estimators (Ledoit and Wolf 2003), applying techniques from robust optimization (Goldfarb and Iyengar 2003, Tütüncü and Koenig 2004) and resampling portfolios (Michaud and Michaud 2008). The same problem in the RP framework has been addressed by some recent works. Kapsos *et al.* (2018) presented three variants of the optimization problem for the RB approach as a potential remedy, where the uncertainty in the covariance matrix was taken into consideration in the framework of robust optimization, and proved that their proposed robust versions are computationally tractable. Costa and Kwon (2019) proposed a factor model of asset returns under a Markov regime-switching framework to more accurately estimate the covariance matrix by adaptively reflecting market conditions. The covariance matrix estimated from the factor model serves as an input to the RP portfolio optimization in an attempt to reduce the sensitivity of the portfolio subject to estimation errors. In addition, they used robust optimization to further improve the portfolio sensitivity by employing uncertainty sets for regression coefficients and residual variances of the factor model. In the work of Savona and Orsini (2019), subsets of the set of all investable assets are considered as a way of diversifying the effect of estimation errors across the subsets. They presented a procedure of clustering RP portfolios for the subsets and combining the clusters into one portfolio in such a way that the merged final portfolio is essentially the tangency portfolio of the clusters.

Our work contributes to the extant literature as follows.

- We investigate the out-of-sample risk contributions of the RP portfolio constituents in the presence of estimation errors in the covariance matrix. Our contribution lies in that the previous studies on the out-of-sample performance of the RP strategy were

mostly about portfolio weights, Sharpe ratios, etc., while leaving the risk contributions of constituents, which are the central concept that this strategy is based on, almost unexplored. Our study reveals that the risk contributions of constituents may perform in a highly unexpected manner in the setting of factor-based investing and clarifies the reason.

- Our work sheds light on the way of using the given data in order to obtain some clue about the out-of-sample performance of risk contributions of constituents. Specifically, we investigate the relationship between specific local information computed from ex ante RP portfolios and the extent to which the risk contributions of constituents are close to each other on an ex post basis.
- We propose an algorithm to construct better RP portfolios in terms of the out-of-sample risk contributions of portfolio constituents. We employ the idea of resampling method of Michaud and Michaud (2008) with an attempt to improve the original algorithm based on the previous findings. We then empirically verify the effectiveness of our proposed method by comparing it with a naive benchmark portfolio strategy and other RP portfolio strategies that take the estimation errors into account.

3. Resampled equal risk contribution portfolios

We consider a universe of p investable assets. We denote the proportions of the investment budget allocated to the assets by a column vector $\mathbf{w} = [w_1, \dots, w_p]^T$ in $\mathbb{R}^p$, identified with a portfolio. Let Σ be the $p \times p$ out-of-sample covariance matrix of the assets' returns. We assume that Σ is positive definite. Recall that the set of all budget-constrained long-only portfolios is represented by a simplex $\Delta^{p-1} = \{\mathbf{w} \in \mathbb{R}^p : \sum_{i=1}^p w_i = 1 \text{ and } w_i \geq 0 \text{ for all } i\}$.

3.1. Equal risk contribution portfolios

We let the volatility of a portfolio's return be a measure of risk, i.e. $\sigma(\mathbf{w}) = \sqrt{\mathbf{w}^T \Sigma \mathbf{w}}$ computes the risk of a portfolio $\mathbf{w} \in \Delta^{p-1}$. Define the risk contribution of asset i as $\sigma_i(\mathbf{w}) = w_i \frac{\partial \sigma}{\partial w_i}(\mathbf{w})$, which yields risk decomposition $\sigma(\mathbf{w}) = \sum_{i=1}^p \sigma_i(\mathbf{w})$. Recall that this way of defining risk contributions of assets is well suited for calculating risk diversification in a financially meaningful way, which is justified by the well-known Euler allocation principle (Tasche 2008). We call $\sigma_i(\mathbf{w})/\sigma(\mathbf{w})$ the percentage risk contribution of asset i . The RP portfolio is defined as the portfolio $\mathbf{w}^* \in \Delta^{p-1}$ that satisfies

$$\sigma_i(\mathbf{w}^*) = \sigma_j(\mathbf{w}^*) \quad \text{for all } i, j \in \{1, \dots, p\}. \quad (1)$$

It can be verified that the following convex optimization problem

$$\min_{\mathbf{w} \in \mathbb{R}^p} \sqrt{\mathbf{w}^T \Sigma \mathbf{w}} \quad \text{subject to} \quad \sum_{i=1}^p \log w_i \geq c \quad (2)$$

with c an arbitrary real number attains a unique solution $\mathbf{y}^*$ in the positive orthant $\mathbb{R}_{++}^p$, which can be found numerically using an off-the-shelf conic programming solver, e.g. SCS (O'Donoghue *et al.* 2016). By virtue of the Karush-Kuhn-Tucker condition, it can be also shown that $\sigma_i(\mathbf{y}^*) = \sigma_j(\mathbf{y}^*)$ for all i, j , leading to the RP portfolio via a normalization $\mathbf{w}^* = \mathbf{y}^* / \sum_{i=1}^p y_i^*$. This leads us to deduce that the RP portfolio always exists and is unique, which can be obtained by solving Problem (2) followed by a normalization. For details, we refer to Kaya and Lee (2012), where the argument in this paragraph is proved for a straightforward equivalent reformation of Problem (2).

Since Σ is unknown *ex ante*, we use its estimate $\hat{\Sigma}$ that is also a symmetric positive definite matrix in $\mathbb{R}^{p \times p}$ computed based on in-sample data. For example, we can use the sample covariance matrix of daily returns over the past one year before a holding period. Replacing Σ with $\hat{\Sigma}$, we define the in-sample RP portfolio as the portfolio $\hat{\mathbf{w}}(\hat{\Sigma}) \in \Delta^{p-1}$, obtained through

$$\min_{\mathbf{w} \in \mathbb{R}^p} \sqrt{\mathbf{w}^\top \hat{\Sigma} \mathbf{w}} \quad \text{subject to} \quad \sum_{i=1}^p \log w_i \geq c. \quad (3)$$

For notational simplicity, we omit $\hat{\Sigma}$ to denote the in-sample RP portfolio if there is no possible confusion. We define the in-sample volatility of a portfolio $\mathbf{w}$ by $\hat{\sigma}(\mathbf{w}) = \sqrt{\mathbf{w}^\top \hat{\Sigma} \mathbf{w}}$ and the in-sample risk contributions and percentage risk contributions similarly to their out-of-sample counterparts.

3.2. Sensitivity of RP portfolios

In this subsection, we argue that the RP approach may not be robust in the sense that the out-of-sample risk contributions are not identical across the portfolio constituents. We note that the sensitivity of risk contributions of RP portfolio's constituents with respect to input parameter variation has been almost unexplored, whereas it is widely known that the weights of an RP portfolio do not vary as much as those of other portfolios as the input parameter to the corresponding optimization problems changes. Indeed, there are mainly two reasons that the weights of a RP portfolio optimized using sample data do not differ much from those constructed out of sample, compared to other portfolio strategies. First, Problem (3) has just one input, the estimated covariance matrix $\hat{\Sigma}$. So, errors in mean estimates are not incorporated in the portfolio selection procedure. This may improve robustness since the optimized portfolio is known to be more vulnerable to errors in mean estimates. Second, in view of Problem (2), the RP portfolio is a constrained, or equivalently regularized, minimum variance portfolio. Specifically, the constraint function $\mathbf{w} \mapsto \sum_{i=1}^p \log w_i$ on the domain $\Delta^{p-1} \cap \mathbb{R}_{++}^p$ attains its maximum at $\mathbf{w} = [1/p, \dots, 1/p]^\top$, implying that it forces sufficient diversification of the optimized portfolio's weights while alleviating the effect of errors in covariance estimates on the portfolio optimization. By construction, however, the main focus of RP portfolios is on risk diversification, which is captured by the concept of risk contributions of portfolio constituents. So, instead of the robustness in the out-of-sample optimized portfolios' weights or the optimal objective value,

we focus on how well the risk contributions of constituents are diversified out of sample.

To investigate, we consider four distinct datasets. The three of them consist of daily returns of six factors – five of them corresponding to Fama-French five factors (Fama and French 2015) and the remaining one to the momentum factor – constructed from US, Europe and Global equity universes. Hereafter, we refer to these six factors as stock factors. For comparison purposes, we also consider another dataset consisting of daily returns of five value-weighted portfolios, each of which represents an industry sector of the US equity universe, where each stock in the universe is assigned to one of the five industry sectors based on the companies' four-digit Standard Industrial Classification code. All of the datasets are available from Kenneth French's data library.[†] Table 1 summarizes some descriptive statistics of the datasets, computed for almost twenty-eight years from November 1991 to September 2019, and figure 1 demonstrates average pairwise correlations between the underlying assets. We can see from table 1 that, compared to the industry sector portfolios, the stock factors are relatively uncorrelated and even have negative correlations for some pairs. It can be observed, either in whole (table 1) or in part (figure 1), that the stock factors are relatively uncorrelated, compared to the industry sectors.

In figure 2, we display 21-day rolling percentage risk contributions of the assets of RP portfolios, formed using sample covariance matrices of daily returns from previous 252 days and held for the following 21 days. As we can see from figure 2(a–c), the out-of-sample percentage risk contributions of the stock factors deviate largely from the expectation that they are close to 16.7%. This suggests the unreliability in estimating the risk contribution that each stock factor actually obtains. This unreliability is because the stock factors have small correlations as demonstrated in table 1 and figure 1, which effectively removes the restriction on percentage risk contributions of portfolio constituents that for $i \in \{1, \dots, p\}$

$$\frac{\sigma_i(\mathbf{w})}{\sigma(\mathbf{w})} = \frac{w_i \sum_{j=1}^p \sigma_{ij} w_j}{\mathbf{w}^\top \Sigma \mathbf{w}} > 0, \quad (4)$$

where σ_{ij} denotes the (i, j) th component of the covariance matrix Σ . This observation is interesting because it is often considered desirable to have investment assets with low correlations for diversifying portfolio risk, whereas our observation implies that this property of having low correlations can be problematic in the RP strategy. Indeed, since the small correlation across factors is likely to have sign changes due to sampling variation, the actual out-of-sample percentage risk contribution $\sigma_i(\hat{\mathbf{w}})$ of the asset i of the in-sample RP portfolio $\hat{\mathbf{w}}$ can be negative, allowing some of the other factors to have unreasonably large percentage risk contributions. This is in contrast to the case where the industry sectors constitute the RP portfolio. Due to the highly positive correlations across all the sectors, the numerator in equation (4) is likely to remain positive in spite of the presence of estimation errors. Furthermore, since the percentage risk contributions always sum to 100%, the actual percentage risk contributions of the

[†] https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html

Table 1. Descriptive statistics of underlying assets (November 1991–September 2019).

	Stock factors (US)						Stock factors (Europe)					
	Mkt-RF	SMB	HML	RMW	CMA	WML	Mkt-RF	SMB	HML	RMW	CMA	WML
Sharpe Ratios	0.42	0.09	0.24	0.55	0.39	0.40	0.27	−0.05	0.51	0.90	0.41	0.84
Annualized Volatilities (%)	17.63	9.08	9.66	7.29	6.61	13.82	17.74	9.46	6.88	4.58	5.19	11.52
Correlations												
Mkt-RF	1.00						1.00					
SMB	0.02	1.00					−0.66	1.00				
HML	−0.06	0.05	1.00				0.12	−0.04	1.00			
RMW	−0.38	−0.30	0.07	1.00			−0.13	0.06	−0.53	1.00		
CMA	−0.35	0.04	0.51	0.24	1.00		−0.30	0.17	0.40	−0.21	1.00	
WML	−0.22	0.00	−0.34	0.14	0.04	1.00	−0.28	0.31	−0.25	0.36	0.06	1.00
	Stock factors (Global)						Industry sectors					
	Mkt-RF	SMB	HML	RMW	CMA	WML	Cnsmr	Manuf	HiTec	Hlth	Other	
Sharpe Ratios	0.36	0.02	0.59	0.97	0.55	0.65	0.52	0.45	0.38	0.42	0.34	
Annualized Volatilities (%)	13.85	6.63	5.82	4.22	4.85	9.87	15.55	17.63	22.70	17.97	21.35	
Correlations												
Mkt-RF	1.00						Cnsmr	1.00				
SMB	−0.54	1.00					Manuf	0.81	1.00			
HML	−0.11	0.11	1.00				HiTec	0.75	0.71	1.00		
RMW	−0.36	0.08	−0.07	1.00			Hlth	0.73	0.67	0.62	1.00	
CMA	−0.37	0.14	0.56	0.10	1.00		Other	0.84	0.81	0.76	0.67	1.00
WML	−0.21	0.24	−0.25	0.22	0.02	1.00						

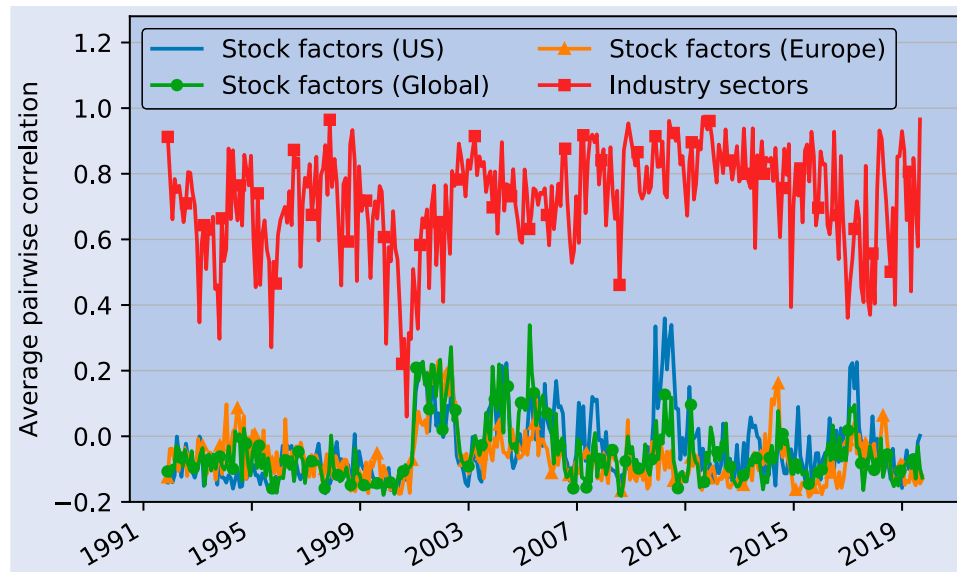


Figure 1. Average 21-day rolling correlations between assets (November 1991–September 2019).

industry sectors are confined in the reasonable range. In fact, the out-of-sample percentage risk contributions of the industry sectors are almost evenly distributed around 20% as shown in figure 2(d).

3.3. Resampling method

3.3.1. Resampling technique in the mean-variance framework. The significant instability of risk contributions of assets observed in figure 2 motivates us to design a way to mitigate the effect of estimation errors to the risk contributions of assets. For this, we consider the resampling technique of Michaud and Michaud (2008) as a potential

remedy. This method was introduced in the context of the classical mean-variance optimization with uncertainty in the means and the covariances of asset returns as a statistical route for furnishing mean-variance optimization with more practical investment value. Here, the process of portfolio optimization are performed multiple times with different input parameters that are sampled from the probability distribution centered at the sample means and the sample covariances. When there is no useful information about the uncertainty in the sampled parameters, all the simulated optimal portfolios are considered to be equally likely, and the method defines the resampled minimum variance portfolio as the average of the portfolio weights of all the simulated optimal portfolios. Since the resampled portfolio is, by construction,

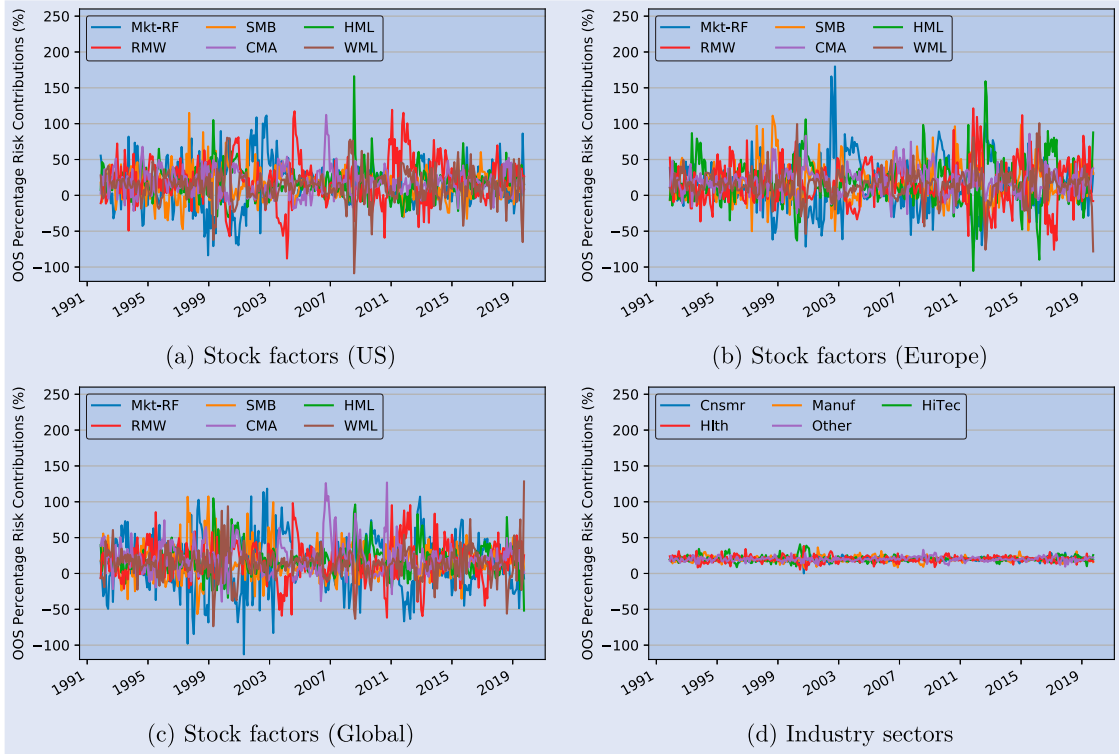


Figure 2. Out-of-sample percentage risk contributions of RP portfolios' constituents. (a) Stock factors (US). (b) Stock factors (Europe). (c) Stock factors (Global) and (d) Industry sectors.

Algorithm 1: Resampled minimum-variance portfolio construction

Input : Previous daily returns $r_{t-1}, \dots, r_{t-T}$ at day t and the number N of resampling

Output: A resampled minimum-variance portfolio $\hat{\mathbf{w}}^*$

- 1 Compute the sample covariance matrix
 $\hat{\Sigma}_{\text{smp}} = \frac{1}{T-1} \sum_{i=1}^T (\mathbf{r}_{t-i} - \bar{\mathbf{r}})(\mathbf{r}_{t-i} - \bar{\mathbf{r}})^\top$, where
 $\bar{\mathbf{r}} = \frac{1}{T} \sum_{i=1}^T \mathbf{r}_{t-i}$
 - 2 **for** $n \leftarrow 1$ **to** N **do**
 - 3 Sample $\hat{\Sigma}_n$ from a distribution centered at $\hat{\Sigma}_{\text{smp}}$.
 - 4 Compute the corresponding minimum-variance portfolio $\hat{\mathbf{w}}_n$.
 - 5 **end**
 - 6 **return** $\hat{\mathbf{w}}^* = \frac{1}{N} \sum_{n=1}^N \hat{\mathbf{w}}_n$
-

not an outlier, it provides reliable investment with better out-of-sample performance on average. The procedure for constructing the resampled minimum-variance portfolio is described in algorithm 1.

3.3.2. Proposed method. We note that there is some clue from which we can evaluate the relevance of the simulated portfolios constructed by the RP strategy, and, in turn, improve the performance of RP portfolios by modifying the original resampling technique of algorithm 1. To make things clear, let us introduce some quantities. First, the Herfindahl index, defined as

$$\tilde{H}(\mathbf{x}) = \sum_{i=1}^p x_i^2 \quad (5)$$

for a vector $\mathbf{x} \in \mathbb{R}^p$ with $\sum_{i=1}^p x_i = 1$, is a measure of diversification often used in the literature.[†] $\tilde{H}(\mathbf{x})$ attains the minimum value if and only if $\mathbf{x}$ is perfectly diversified, i.e. $\mathbf{x} = [1/p, \dots, 1/p]^\top$, and it increases as $\mathbf{x}$ becomes more poorly diversified. Let an in-sample covariance matrix and its corresponding RP portfolio be denoted by $\hat{\Sigma}$ and $\hat{\mathbf{w}}$, respectively. Then, the diversification of the in-sample risk contributions is computed by $H : \Delta^{p-1} \rightarrow \mathbb{R}$ defined as

$$H(\mathbf{w}; \hat{\Sigma}) = \tilde{H} \left(\frac{\hat{\sigma}_1(\mathbf{w})}{\hat{\sigma}(\mathbf{w})}, \dots, \frac{\hat{\sigma}_p(\mathbf{w})}{\hat{\sigma}(\mathbf{w})} \right) \text{ for a portfolio } \mathbf{w} \in \Delta^{p-1}, \quad (6)$$

where $\hat{\sigma}$ and $\hat{\sigma}_i$ are associated with $\hat{\Sigma}$.

Using this notion, we can investigate how well the risk contributions of portfolio constituents are diversified. We first note that the function H may change abruptly near the out-of-sample RP portfolio $\mathbf{w}^*$ as presented in figure 3 in which for illustrative purposes we consider a simplified setting where just three stock factors (Mkt-RF, HML and SMB) constitute portfolios. It shows that tiny variations in $\mathbf{w}^*$ can lead to highly concentrated risk contributions of assets. Indeed, even if the covariance estimates have large errors, the in-sample RP portfolio is quite close to the true one since both are constrained to a convex subset of Δ^{p-1} centered at $[1/p, \dots, 1/p]^\top$. Thus, we can deduce that the actual risk contributions of assets may become concentrated because of the change of the landscape rather than the change of portfolio weights. In particular, the local property of H plays a

[†] The Herfindahl index is usually defined for vectors in Δ^{p-1} . We extend $\tilde{H}$ in that it is allowed to compute vectors not in the non-negative orthant $\mathbb{R}_+^p$ so as to measure the diversification of risk contributions of assets with negative values.

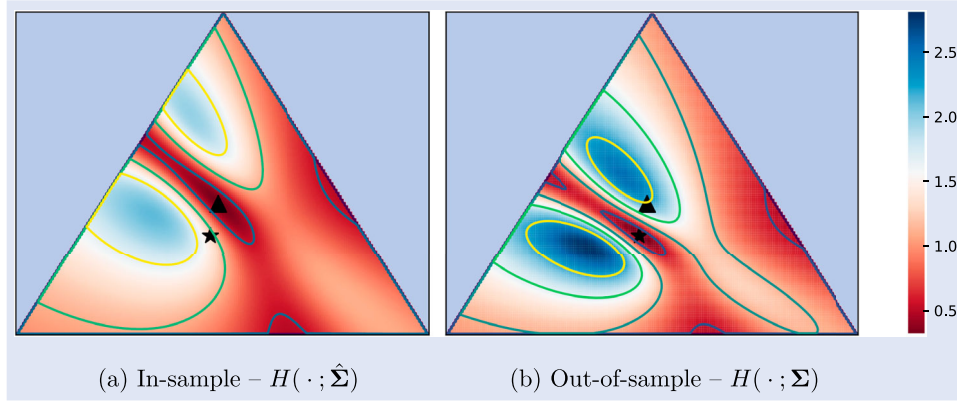


Figure 3. Landscapes of Herfindahl index H of risk contributions of portfolios' constituents on the simplex Δ^2 of all portfolios composed of the Fama-French three factors in an investment period. Here, each vertex corresponds to a portfolio consisting of a single factor. The triangle and star on the simplex correspond to the in-sample and the out-of-sample RP portfolios, respectively. (a) In-sample $- H(\cdot; \hat{\Sigma})$ and (b) Out-of-sample $- H(\cdot; \Sigma)$.

key role in the deterioration. Namely, if $H(\cdot; \Sigma)$ changes abruptly near $\mathbf{w}^*$ in some direction, the in-sample RP portfolio $\hat{\mathbf{w}}$ is likely to have poor out-of-sample performance, i.e. large $H(\hat{\mathbf{w}}; \Sigma)$. We characterize this local structure by a ratio κ of the largest eigenvalue to the smallest eigenvalue of the Hessian matrix $\nabla^2 H(\mathbf{w}^*; \Sigma)$ projected on Δ^{p-1} , often referred to as the condition number, which effectively measures eccentricity of level sets of $H(\cdot; \Sigma)$ near $\mathbf{w}^*$ on Δ^{p-1} . With slight abuse of notation, we call it the condition number at $(\mathbf{w}^*, \Sigma)$.

Without knowing Σ , however, we cannot take advantage of the local information. Instead, we use the in-sample information of $\hat{\Sigma}$ and $\hat{\mathbf{w}}$ in a natural way by employing the condition number $\hat{\kappa}$ at $(\hat{\mathbf{w}}, \hat{\Sigma})$ as a proxy for κ . We then consider a simulated portfolio $\hat{\mathbf{w}}_n$ to be relevant if its corresponding condition number $\hat{\kappa}_n$ is small. We study the validity of the aforementioned arguments via statistical analysis in Section 3.4.

Based on this idea, we propose algorithm 2. As in the original resampling technique, we generate Monte Carlo simulated RP portfolios $\hat{\mathbf{w}}_1, \dots, \hat{\mathbf{w}}_N$. (We fix the distribution of sample covariance matrices to be the Wishart distribution.) Then, we define the resampled RP portfolio $\hat{\mathbf{w}}^*$ as the weighted average of the simulated RP portfolios. For each simulated covariance matrix $\hat{\Sigma}_n$ and its corresponding RP portfolio $\hat{\mathbf{w}}_n$, we let the weight θ_n be inversely proportional to the condition number $\hat{\kappa}_n$ at $(\hat{\mathbf{w}}_n, \hat{\Sigma}_n)$, i.e. the ratio of the smallest eigenvalue to the largest eigenvalue of the Hessian matrix $\nabla^2 H(\hat{\mathbf{w}}_n; \hat{\Sigma}_n)$ projected on Δ^{p-1} whose eigenvalues are nonnegative due to Proposition 3.1. By doing so, we weigh more on the simulated RP portfolios that are likely to work better based on the local structure of the landscape of H to obtain our final resampled RP portfolio $\hat{\mathbf{w}}^*$.

PROPOSITION 3.1 *The in-sample RP portfolio $\hat{\mathbf{w}}$ is the unique minimizer of $H(\cdot; \hat{\Sigma})$ in equation (6). Moreover, the Hessian $\nabla^2 H(\hat{\mathbf{w}}; \hat{\Sigma}) \in \mathbb{R}^{(p-1) \times (p-1)}$ on Δ^{p-1} is positive semidefinite.*

Proof Clearly, $H(\cdot; \hat{\Sigma})$ is continuous on the compact set Δ^{p-1} , implying that it attains the minimum value. Furthermore, for the Herfindahl index $\tilde{H}$ defined on $\{\mathbf{x} \in \mathbb{R}^p :$

Algorithm 2: Resampled RP portfolio construction

Input : Previous daily returns $r_{t-1}, \dots, r_{t-T}$ at day t and the number N of resampling

Output: A resampled RP portfolio $\hat{\mathbf{w}}^*$

- 1 Compute the sample covariance matrix $\hat{\Sigma}_{\text{smp}} = \frac{1}{T-1} \sum_{i=1}^T (\mathbf{r}_{t-i} - \bar{\mathbf{r}})(\mathbf{r}_{t-i} - \bar{\mathbf{r}})^\top$, where $\bar{\mathbf{r}} = \frac{1}{T} \sum_{i=1}^T \mathbf{r}_{t-i}$
 - 2 **for** $n \leftarrow 1$ **to** N **do**
 - 3 Sample $\hat{\Sigma}_n$ from a distribution centered at $\hat{\Sigma}_{\text{smp}}$.
 - 4 Compute the corresponding RP portfolio $\hat{\mathbf{w}}_n$ and weight θ_n .
 - 5 **end**
 - 6 **return** $\hat{\mathbf{w}}^* = \sum_{n=1}^N \theta_n \hat{\mathbf{w}}_n$
-

$\sum_{i=1}^p x_i = 1\}$, $\tilde{H}(\mathbf{x}^*)$ attains the minimum value if and only if

$$x_i^* = \frac{1}{p} \quad \text{for all } i. \quad (7)$$

Since the in-sample RP portfolio $\hat{\mathbf{w}}$ is the unique portfolio in Δ^{p-1} satisfying

$$\frac{\hat{\sigma}_i(\mathbf{w})}{\hat{\sigma}(\mathbf{w})} = \frac{1}{p} \quad \text{for all } i,$$

it is the unique minimizer of $H(\cdot; \hat{\Sigma})$.

Next, we may extend the function $H(\cdot; \hat{\Sigma})$ to the set $S = \{\mathbf{w} \in \mathbb{R}^p : \sum_{i=1}^p w_i = 1\}$. It can be easily verified that there exists a bijective function $f : S \rightarrow \mathbb{R}^{p-1}$ defined by

$$f(\mathbf{w}) = \mathbf{A}\mathbf{w} + \mathbf{b} \quad \text{for } \mathbf{w} \in S,$$

where $\mathbf{A} \in \mathbb{R}^{(p-1) \times p}$ and $\mathbf{b} \in \mathbb{R}^{p-1}$. Using f , we define a function $G : \mathbb{R}^{p-1} \rightarrow \mathbb{R}$ by setting

$$G(\mathbf{y}) = H(\mathbf{w}; \hat{\Sigma})$$

for all $\mathbf{y} \in \mathbb{R}^{p-1}$ with $\mathbf{y} = f(\mathbf{w})$. Now, let $\hat{\mathbf{y}} = f(\hat{\mathbf{w}})$. Since $\hat{\sigma}_1(\hat{\mathbf{w}})/\hat{\sigma}(\hat{\mathbf{w}}), \dots, \hat{\sigma}_p(\hat{\mathbf{w}})/\hat{\sigma}(\hat{\mathbf{w}})$ satisfy equation (7), $\hat{\mathbf{y}}$ minimizes the function G . Due to the second-order condition for a

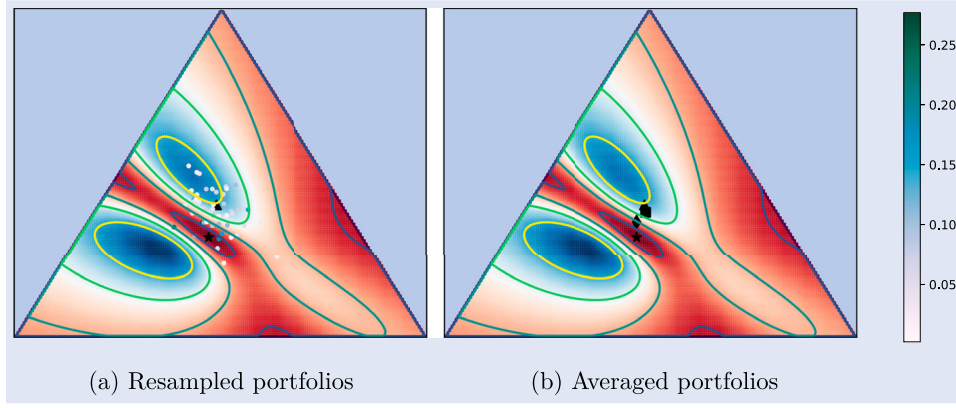


Figure 4. Illustration of the way algorithm 1 works on the landscape of H that is the same with figure 3(b). The triangle and star represent the RP portfolios as in figure 3. The diamond and square correspond to RP-NR and ours, respectively. Color bar on the right represent the intensity of a weight of each resampled portfolio. (a) Resampled portfolios and (b) Averaged portfolios.

minimum, this implies that the matrix $\nabla^2 G(\hat{\mathbf{y}}) \in \mathbb{R}^{(p-1) \times (p-1)}$ is positive semi-definite. Thus, for all $\mathbf{d} \in \mathbb{R}^p$ parallel to the set S ,

$$\mathbf{d}^\top \nabla^2 H(\hat{\mathbf{w}}; \hat{\Sigma}) \mathbf{d} = \mathbf{d}^\top \mathbf{A}^\top \nabla^2 G(\mathbf{A}\hat{\mathbf{w}} + \mathbf{b}) \mathbf{A} \mathbf{d} \geq 0,$$

where the equality follows from simple matrix calculus. ■

Figure 4 illustrates the way algorithm 2 performs. First, the simulated RP portfolios $\hat{\mathbf{w}}_1, \dots, \hat{\mathbf{w}}_N$, represented by small dots, and their corresponding weights $\theta_1, \dots, \theta_N$, represented by the color on the dots, are obtained as shown in figure 4(a). We can see that the simulated RP portfolios are widely distributed around the in-sample RP portfolio. When they are averaged with equal weights, the resulting portfolios, represented by the square in figure 4(b), cannot be located far enough from the in-sample RP portfolio. By computing weighted average, on the other hand, the simulated portfolios that are likely to poorly perform are properly diminished so that the resulting portfolio approaches the out-of-sample RP portfolio.

Our proposed portfolio optimization can be interpreted in the view of imposing a norm constraint to the portfolio weights of the RP optimization problem (3), but with an induced bias towards the portfolio with the out-of-sample risk contributions being equal. To be more clear, if we define a noise matrix $\mathcal{E}_n := \hat{\Sigma}_n - \hat{\Sigma}$, then the resampled portfolio $\hat{\mathbf{w}}_n$ is the solution of the optimization problem

$$\min_{\mathbf{w} \in \mathbb{R}^p} \sqrt{\mathbf{w}^\top \hat{\Sigma} \mathbf{w}} \quad \text{subject to} \quad \sum_{i=1}^p \log w_i \geq c \quad \text{and} \quad \sqrt{\mathbf{w}^\top \mathcal{E}_n \mathbf{w}} \leq \delta, \quad (8)$$

which is equivalent to Problem (3) with additional constraint that

$$\sqrt{\mathbf{w}^\top \mathcal{E}_n \mathbf{w}} \leq \delta \quad (9)$$

for some $\delta > 0$, which looks similar to the matrix norm-constrained portfolio optimization studied by DeMiguel *et al.* (2009a). Since $\mathcal{E}_n$ is not necessarily positive definite, it does not define a norm. With resamples, however, our method imposes the constraint (9) multiple times to the problem,

Table 2. Average Herfindahl index of various RP portfolios over sample period of November 1991–September 2019.

	US	Europe	Global
RP	0.5282	0.5951	0.5611
RP-NR	0.5245	0.5926	0.5584
Ours	0.5051	0.5761	0.5514

essentially leading to 2-norm constrained RP portfolios when algorithm 1 is conducted. Algorithm 2 then induces the bias on the portfolio weights towards better risk contributions by computing weighted average.

To see how the proposed method generates the induced bias, we computed values of the function H for the datasets described in Section 3.2, and report their average over all the holding periods in table 2. Here, we compare our algorithm with the conventional RP portfolios and resampled portfolio constructed through equally weighted averaging, i.e. we set θ_n in algorithm 2 to $1/N$ for all n . This amounts to naively applying the resampling method of Michaud and Michaud (2008) to the RP approach without using the local information captured by the condition numbers $\hat{\kappa}_n$ of the Herfindahl index. We refer to it as (RP-NR). Since the resampling-based RP portfolios are stochastically constructed, the experimental results reported for RP-NR and our proposed method are the averages of 100 runs. We can see from the table that both of the resampling-based methods reduce the concentration of risk contributions of portfolio constituents. We also note that our method does improve over RP-NR for all the three cases.

3.4. Relationship between concentration of risk contributions and local structure

In this subsection, we examine the relationship between the concentration of risk contributions of assets and the local information near in-sample and out-of-sample RP portfolios via regression analysis. We regress the out-of-sample Herfindahl indexes $H(\hat{\mathbf{w}}; \Sigma)$ of in-sample RP portfolios $\hat{\mathbf{w}}$ on the condition number κ at $(\mathbf{w}, \Sigma)$, which we call the out-of-sample condition number (Panel A of table 3), and the

Table 3. The relationship between risk contributions' concentration and the out-of-sample and the in-sample condition numbers.

	coefficient (t -statistic)	standard error	Adjusted R^2
Panel A: Regression on out-of-sample condition numbers			
US	3.917×10^{-4} (15.611)	0.251×10^{-4}	0.422
Europe	1.902×10^{-4} (13.679)	0.139×10^{-4}	0.350
Global	1.368×10^{-4} (7.573)	0.181×10^{-4}	0.140
Panel B: Regression on in-sample condition numbers			
US	9.922×10^{-4} (3.280)	3.025×10^{-4}	0.028
Europe	10.966×10^{-4} (8.529)	1.286×10^{-4}	0.172
Global	8.954×10^{-4} (5.098)	1.756×10^{-4}	0.067

condition number $\hat{\kappa}$ at $(\hat{\mathbf{w}}, \hat{\Sigma})$, which we call the in-sample condition number (Panel B of table 3).

The positive t -statistics indicate the positive dependence between the out-of-sample Herfindahl index and the condition numbers, which is consistent with our insight based on figure 3. All of the associated p values of the reported t -statistics in table 3 are less than 0.002, and thus indicate that the dependence is statistically significant. Comparing the t -statistics between out-of-sample and in-sample condition numbers, we can see that the association with the Herfindahl index becomes lower when the in-sample condition numbers are independent variable of the regression analysis, but there still is the association with high statistical significance. Even if the adjusted R^2 appear small in Panel B, the regressions are still significant as the t -statistics indicate.

4. Empirical study

We first carry out an out-of-sample analysis based on datasets of historical daily returns of the six stock factors – the market factor (Mkt-RF), value factor (HML), size factor (SMB), profitability factor (RMW), investment factor (CMA) and momentum factor (WML) – for the three different regions– US, Europe and Global – as we did in Section 3. We estimate the out-of-sample covariance matrix over a holding period of 21 days (one trading month) using the factors' daily returns and a rolling window period of 252 days (one trading year). The evaluations are performed for three different sample periods. (i) First, we consider the entire sample period: from July 1963 to November 2019 for the US dataset and from November 1990 to September 2019 for the datasets of other regions. These consist of 664 and 347 non-overlapping holding periods for the US and for the other regions, respectively. In order to see the performance under stressed market conditions, the evaluation was also carried out over the following sample periods: (ii) a sample period from January 2006 to December 2010 and (iii) a sample period from January 1998 to December 2002. The periods of (ii) and (iii) cover five years around the 2007–2008 global financial crisis and the dot-com bubble burst of the early 2000s, respectively, both of which amount to 60 iterative performance evaluations. The experimental results obtained for RP-NS and our proposed method are the average of 100 runs.

All the experiments were performed on a personal computer with Inter Core i7-8700 K 3.70 GHz and 32 GB DDR4 RAM running Ubuntu 16.04.3 LTS. The considered algorithms were implemented in Python (version 3.6.4) using an optimization modeling language CVXPY (version 0.4.11) (Diamond and Boyd 2016, Agrawal *et al.* 2018) with SCS (version 2.0.2) (O'Donoghue *et al.* 2016) as a solver.

4.1. Competing strategies

We compare the out-of-sample performance of our method to those of the following portfolio strategies. (i) The equally weighted portfolio (EW) strategy has the portfolio weight $\mathbf{w}_{EW} = [1/p, \dots, 1/p]^\top$. This strategy has been widely used in the literature as a benchmark because it often shows a comparable performance to various sophisticated portfolio strategies as reported in DeMiguel *et al.* (2009b). (ii) We consider the standard RP portfolio explained in Section 3.1, which can be obtained by solving Problem (3) with the sample covariance matrix $\hat{\Sigma}_{\text{smpl}}$ used as the input parameter. We refer to it as RP. We also consider three different methods devised to reduce the impact of estimation errors in RP portfolios: (iii) We evaluate the RP portfolio constructed using a shrinkage covariance estimate proposed in Ledoit and Wolf (2003) as an input parameter to Problem (3). We refer to it as RP-Sh. To use a shrinkage covariance estimate is a typical way of reducing the impact of estimation errors in covariance estimates on the optimized portfolio, which is interpreted as applying a structural assumption on asset returns. (iv) The robust risk budgeting (RRB) method of Kapsos *et al.* (2018) was proposed as an alternative approach to risk budgeting method for mitigating estimation errors in covariance estimates in the framework of robust optimization. The RRB method is formulated by the optimization problem

$$\min_{\mathbf{w} \in \mathbb{R}^p} \max_{\Sigma \in \mathcal{S}} \mathbf{w}^\top \Sigma \mathbf{w} - \sum_{i=1}^p b_i \log w_i \quad \text{subject to} \quad \sum_{i=1}^p w_i = 1, \quad (10)$$

where $b_1, \dots, b_p$ are risk budgets and $\mathcal{S}$ is an uncertainty set. In our study, we fix each of the risk budgets b_i of portfolio constituents to be equal, where the problem reduces to the robust version of the RP approach, and we consider two forms of uncertainty mentioned in Kapsos *et al.* (2018). When $\mathcal{S}$ is finite, denoted by RRB-f, for each holding period, we consider 11 different sets consisting of a sample covariance matrix of daily returns over the entire formation period of $k \in \{3, 4, 6, 12\}$ months and an l number of sample covariance matrices of consecutive daily returns over the non-overlapping periods of $k/l \in \mathbb{N}$ months in the formation period. Having observed the out-of-sample performance for the candidate uncertainty sets, we chose the one that performs best in risk contributions' concentration and Sharpe ratio most frequently over the datasets and time periods. When $\mathcal{S}$ is infinite, denoted by RRB-i, it is defined as

$$\mathcal{S} = \{ \Sigma \in \mathbb{R}^{p \times p} : \Sigma^l \leq \Sigma \leq \Sigma^u, \\ \Sigma \text{ is positive semidefinite.} \},$$

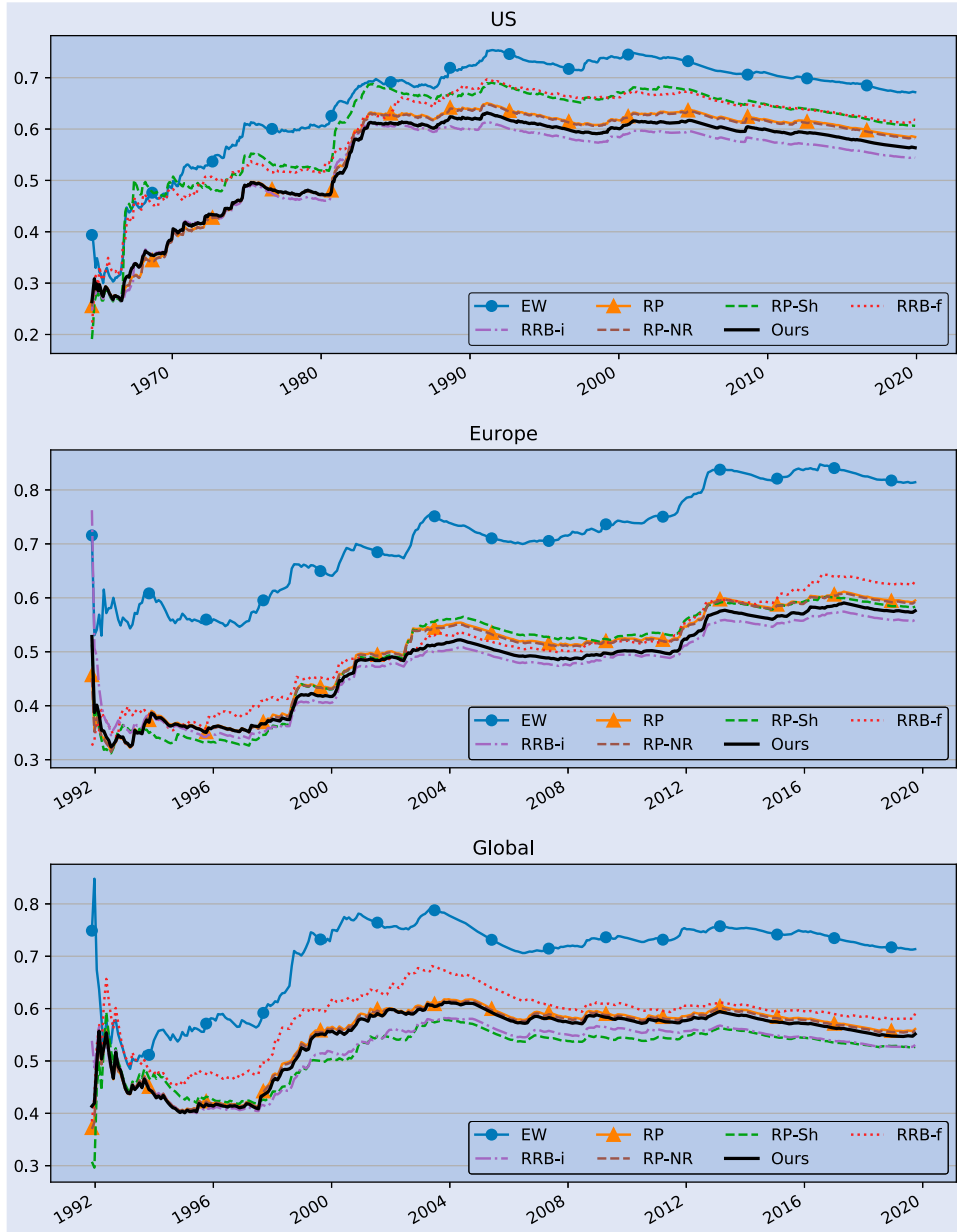


Figure 5. Cumulatively averaged Herfindahl index over the entire sample period

where the inequalities are element-wise. The bounds Σ^l and Σ^u are constructed by using the standard error of the sample covariance matrix $\hat{\Sigma}$. Specifically, we let $\sigma_{ij}^l = \hat{\sigma}_{ij} - \alpha s_{ij} \times \hat{\sigma}_{ij}$ and $\sigma_{ij}^u = \hat{\sigma}_{ij} + \alpha s_{ij} \times \hat{\sigma}_{ij}$, for $\alpha \in \{1.0, 2.0, 3.0, 4.0\}$, and choose the best α as in RRB-f. Here, $\hat{\sigma}_{ij}$, σ_{ij}^l , σ_{ij}^u and s_{ij} are the (i, j) -th components of $\hat{\Sigma}$, Σ^l , Σ^u and the standard error of $\hat{\Sigma}$, respectively. (v) Finally, we evaluate the performance of the naively resampled RP portfolio (RP-NR) that was explained in Section 3.3.

4.2. Out-of-Sample performance evaluations

Now, we present the out-of-sample test results. We first investigate the diversification of the out-of-sample risk contributions across portfolio constituents by calculating the Herfindahl indexes $H(\hat{\mathbf{w}}_t; \Sigma_t)$, where t denotes the time index of holding periods, $\hat{\mathbf{w}}_t$ is a RP portfolio constructed before the

beginning of the t -th holding period, and Σ_t is the covariance matrix on the t -th holding period. Recall that the smaller the value of H , the more diversified are the risk contributions. Figure 5 exhibits curves of H values averaged in a cumulative manner, i.e. a point of a curve at time t represents

$$\frac{1}{t} \sum_{\tau=1}^t H(\hat{\mathbf{w}}_{\tau}; \Sigma_{\tau}).$$

We interpret the results in two respects. First, the resampling-based portfolios always outperform the conventional RP portfolio. For example, our proposed method reduces the H value, averaged over the entire period, of the conventional method by 3.6%, 3.2% and 1.7% for the US, Europe and Global datasets, respectively. This consistent outperformance is attributed to its nature of eliminating outliers by merging a multiple of simulated RP portfolios. Also, we can say that

the difference of the resampling-based methods to the conventional RP method is less dramatic than the other approaches. Since the amount to which our method improves the out-of-sample performance is relatively small, it sometimes occurs that other portfolio outperforms ours, e.g. the H value of RP-Sh on the Global dataset. From the perspective of consistency, however, ours significantly outperforms other modified RP methods.

Second, we also note that our method outperforms RP-NR in all the cases. Figure 5 shows that the concentration of the out-of-sample risk contributions of RP-NR is almost identical to that of RP. This is expected from figure 4 where the RP-NR is closer to RP than the resampled portfolio with the proposed weighting scheme. We believe that the almost identical behavior of RP-NR and RP can also be explained by the nature of unbiasedness of the resampled covariance matrices sampled from the Wishart distribution which will produce resampled portfolio weights consistent with those of RP as was proved for various optimized portfolio weights in the mean-variance framework (Okhrin and Schmid 2006). The resampled RP portfolio with our proposed weighting scheme, on the other hand, benefits from the induced bias in that the concentration of the out-of-sample risk contributions is reduced. This suggests the effectiveness of using the information captured by the in-sample condition numbers instead of naively averaging the simulated portfolios.

To see the performance of the aforementioned portfolio strategies in terms of traditional performance measures, we exhibit annualized returns, annualized volatilities and Sharpe ratios with their corresponding statistical test results in tables 4–6. We computed the Sharpe ratio as the ratio of the annualized portfolio return to the annualized volatility of the portfolio returns. Note that we do not subtract the risk-free rate from the return when computing Sharpe ratios since the considered stock factors except for the market factor are derived from self-financing long/short portfolios, which, in effect, net out the returns due to the risk-free rate. As for the market factor, its returns are already subtracted from the risk-free rate. To evaluate the out-of-sample performance of the aforementioned portfolio strategies, we also compute two-sided p -values for the hypothesis that the Sharpe ratio of each portfolio strategy i is equal to that of RP-Sh that is chosen as the one that each strategy is compared to, given the popularity of the shrinkage method. We use the studentized circular block bootstrapping method proposed by Ledoit and Wolf (2008) with $B = 1,000$ bootstrap resamples and a block size $b = 5$. We use the code available at <https://www.econ.uzh.ch/en/people/faculty/wolf>.

Tables 4–6 indicate that the Sharpe ratio of our method is not statistically significantly different from that of RP-Sh. At the same time, however, we note that the out-of-sample Sharpe ratio of our proposed portfolio has consistent performances, outperforming over RP uniformly and over RP-Sh uniformly except for one case. This is similar to what we have observed in the out-of-sample performance of risk contributions of portfolio constituents. This is further shown in table 7 where we summarize the performance comparison of Sharpe ratios by giving each portfolio strategy a score from the range of 1 to 7. A larger score represents a higher Sharpe ratio. It is

Table 4. Performance measures for the entire sample period.

	Annualized Return	Annualized Volatility	Sharpe Ratio
Panel A: US			
EW	0.0455	0.0374	1.2166 (0.28)
RP	0.0373	0.0282	1.3227 (0.95)
RP-Sh	0.0417	0.0314	1.3280 (1.00)
RRB-f	0.0361	0.0277	1.3032 (0.79)
RRB-i	0.0392	0.0284	1.3803 (0.30)
RP-NR	0.0374	0.0282	1.3262 (0.94)
Ours	0.0375	0.0281	1.3345 (0.88)
Panel B: Europe			
EW	0.0445	0.0353	1.2606 (0.13)
RP	0.0369	0.0241	1.5311 (0.05)
RP-Sh	0.0370	0.0260	1.4231 (1.00)
RRB-f	0.0344	0.0223	1.5426 (0.31)
RRB-i	0.0371	0.0242	1.5331 (0.05)
RP-NR	0.0369	0.0241	1.5311 (0.05)
Ours	0.0372	0.0240	1.5500 (0.03)
Panel C: Global			
EW	0.0403	0.0324	1.2438 (0.21)
RP	0.0349	0.0235	1.4851 (0.22)
RP-Sh	0.0348	0.0245	1.4204 (1.00)
RRB-f	0.0317	0.0213	1.4883 (0.56)
RRB-i	0.0348	0.0238	1.4622 (0.32)
RP-NR	0.0349	0.0235	1.4851 (0.24)
Ours	0.0349	0.0234	1.4915 (0.20)

The p -values, written in the parentheses, are for the hypothesis that the Sharpe ratio for a strategy is different from that for RP-Sh.

exhibited in the table that our method performs consistently well, producing at least a score of 5. Among RP strategies, our method performs at least the second best in most cases.

4.3. Performance in practice

We now consider exchanged traded funds (ETFs) listed in table 8 over the period of July 19th 2013 to April 30th 2020 to see how our method performs in practice. Our study is based on a rolling window analysis with the same size of formation and holding periods used in the previous section, which amounts to 72 non-overlapping holding periods. The ETFs are designed to track market indexes or to provide exposure to various stock factors in the US stock market. They are traded in the New York Stock Exchange and are considered as a practical way of diversifying the portfolio across different risk factors. Since some of them are rather new compared to other traditional ETFs, trading volumes can be smaller causing higher transaction costs (TC). We take this into account by employing the method often used in the literature, e.g. DeMiguel *et al.* (2013), Clark *et al.* (2020). Specifically, we compute the time series of TC-adjusted returns $\tilde{r}_{i,t}$ of each portfolio strategy i by penalizing the portfolio returns $r_{i,t}$ as

$$\tilde{r}_{i,t+1} = r_{i,t+1} - \text{TC} \times \|\mathbf{w}_{t+1} - \mathbf{w}_t \circ (\mathbf{1} + R_{t+1})\|_1, \quad (11)$$

where $\circ$ is the element-wise product, R_{t+1} is the vector of returns of portfolio constituents between t and $t + 1$, $\mathbf{w}_t$ is the vector of weights at time t , $\mathbf{1}$ is the vector of ones, and TC is an

Table 5. Performance measures for sample period of January 2006–December 2010.

	Annualized Return	Annualized Volatility	Sharpe Ratio
Panel A: US			
EW	0.0146	0.0453	0.3223 (0.31)
RP	0.0177	0.0349	0.5072 (0.25)
RP-Sh	0.0214	0.0335	0.6388 (1.00)
RRB-f	0.0156	0.0354	0.4407 (0.18)
RRB-i	0.0209	0.0333	0.6276 (0.87)
RP-NR	0.0177	0.0349	0.5072 (0.27)
Ours	0.0185	0.0345	0.5362 (0.38)
Panel B: Europe			
EW	0.0298	0.0371	0.8032 (0.39)
RP	0.0264	0.0237	1.1139 (0.59)
RP-Sh	0.0249	0.0241	1.0332 (1.00)
RRB-f	0.0222	0.0211	1.0521 (0.95)
RRB-i	0.0266	0.0226	1.1770 (0.49)
RP-NR	0.0264	0.0237	1.1139 (0.59)
Ours	0.0269	0.0234	1.1496 (0.48)
Panel C: Global			
EW	0.0188	0.0336	0.5595 (0.39)
RP	0.0222	0.0233	0.9528 (0.68)
RP-Sh	0.0205	0.0233	0.8798 (1.00)
RRB-f	0.0207	0.0208	0.9952 (0.78)
RRB-i	0.0210	0.0228	0.9211 (0.79)
RP-NR	0.0221	0.0233	0.9485 (0.67)
Ours	0.0225	0.0231	0.9740 (0.58)

The p -values, written in the parentheses, are for the hypothesis that the Sharpe ratio for a strategy is different from that for RP-Sh.

Table 6. Performance measures for sample period of January 1998–December 2002.

	Annualized Return	Annualized Volatility	Sharpe Ratio
Panel A: US			
EW	0.0787	0.0559	1.4079 (0.38)
RP	0.0590	0.0493	1.1968 (0.68)
RP-Sh	0.0550	0.0500	1.1000 (1.00)
RRB-f	0.0434	0.0470	0.9234 (0.57)
RRB-i	0.0641	0.0508	1.2618 (0.37)
RP-NR	0.0592	0.0493	1.2008 (0.64)
Ours	0.0602	0.0494	1.2186 (0.63)
Panel B: Europe			
EW	0.0675	0.0548	1.2318 (0.72)
RP	0.0467	0.0387	1.2067 (0.43)
RP-Sh	0.0458	0.0405	1.1309 (1.00)
RRB-f	0.0471	0.0357	1.3193 (0.47)
RRB-i	0.0482	0.0398	1.2111 (0.31)
RP-NR	0.0468	0.0387	1.2093 (0.44)
Ours	0.0476	0.0390	1.2205 (0.33)
Panel C: Global			
EW	0.0717	0.0526	1.3631 (0.71)
RP	0.0511	0.0395	1.2937 (0.53)
RP-Sh	0.0486	0.0398	1.2211 (1.00)
RRB-f	0.0430	0.0345	1.2464 (0.90)
RRB-i	0.0528	0.0401	1.3167 (0.40)
RP-NR	0.0513	0.0395	1.2987 (0.53)
Ours	0.0516	0.0393	1.3130 (0.43)

The p -values, written in the parentheses, are for the hypothesis that the Sharpe ratio for a strategy is different from that for RP-Sh.

arbitrary transaction cost. Given the possibility of relatively low trading volume of some of the factor ETFs investigated, we set TC to 100 basis points to compute the TC-adjusted returns for our analysis.

We present the performance measures of the portfolio strategies in table 9. As in the previous section, we computed three conventional performance measures as well as p -values for the hypothesis that Sharpe ratio of a portfolio strategy is different from that of RP-Sh, using the method of Ledoit and Wolf (2008). Furthermore, the last column of table 9 lists the turnover of the portfolios that is defined as

$$\text{turnover} = \frac{1}{T-1} \sum_{t=1}^{T-1} \|\mathbf{w}_{t+1} - \mathbf{w}_t \circ (\mathbf{1} + R_{t+1})\|_1. \quad (12)$$

We computed the Sharpe ratio as the ratio of the annualized portfolio return to the annualized volatility of the TC-adjusted portfolio returns. The risk-free rate used to compute annualized excess returns are the one-month Treasury bill rate available at the Kenneth French's data library. We can see from the table that our proposed method performs the best in Sharpe ratio with statistical significance comparable to other strategies. We also note that the smallest turnover incurred by our method among the RP strategies directs to the practical benefit of our method. This is further supported by the observation that the extent to which the Sharpe ratios of our method exceeds others is greater than the results on the six stock factors shown in the previous section, where transaction costs were not taken into consideration in the performance evaluations.

Table 7. Performance comparison among the portfolio strategies. (Larger score indicates higher Sharpe ratio; 7 (best), 1 (worst)).

	Entire period			2006-2010			1998-2002		
	US	Europe	Global	US	Europe	Global	US	Europe	Global
EW	1	1	1	1	1	1	7	6	7
RP	3	4	5	3	5	5	3	2	3
RP-Sh	5	2	2	7	2	2	2	1	1
RRB-f	2	6	6	2	3	7	1	7	2
RRB-i	7	5	3	6	7	3	6	4	6
RP-NR	4	3	4	4	4	4	4	3	4
Ours	6	7	7	5	6	6	5	5	5

Table 8. List of factor ETFs.

Symbol	ETF Name	Factor Tilted Toward	Inception Date
SPY	SPDR S&P 500 ETF	Market	01/22/1993
SIZE	iShares Edge MSCI USA Size Factor ETF	Size	04/16/2013
CHEP	AGFiQ US Market Neutral Value Fund	Value	09/13/2011
MMTM	SPDR S&P 1500 Momentum Tilt ETF	Momentum	10/24/2012
SPHQ	Invesco S&P 500 Quality ETF	Profitability	12/06/2005
BTAL	AGFiQ U.S. Market Neutral Anti-Beta Fund	Beta	09/13/2011

Table 9. Performance measures for the factor ETFs.

	Annualized Excess Return	Annualized Volatility	Sharpe Ratio	Turnover
EW	0.0477	0.0889	0.5367 (0.97)	0.0305
RP	0.0275	0.0581	0.4733 (0.43)	0.0651
RP-Sh	0.0319	0.0590	0.5409 (1.00)	0.0634
RRB-f	0.0200	0.0573	0.3493 (0.28)	0.1650
RRB-i	0.0296	0.0613	0.4831 (0.20)	0.0641
RP-NR	0.0288	0.0576	0.5003 (0.65)	0.0645
Ours	0.0311	0.0561	0.5544 (0.73)	0.0607

The p -values, written in the parentheses, are for the hypothesis that the Sharpe ratio for a strategy is different from that for RP-Sh.

5. Conclusion

This paper presents the risk parity strategy under uncertainty for construction of a portfolio composed of stock factors. We explore the problem of sensitivity of the risk parity portfolio to estimation errors in the covariance matrix and observe that the optimized portfolio may not work as expected from the perspective of risk contributions of portfolio constituents. Specifically, we show that the risk parity approach, which is considered to be robust in the sense that its weights do not vary rapidly as the covariance estimates changes, do not necessarily work well in the presence of estimation errors in that the out-of-sample risk contributions of constituents may not be equal.

An intuitive explanation of the sensitivity of risk contributions of constituents to estimation errors is given by investigating landscape of the Herfindahl index of risk contributions in the simplex that portfolios lie on. A regression analysis verifies the intuition by demonstrating a statistically significant association between the sensitivity and the local structure of the Herfindahl index near the risk parity portfolio. Using this fact upon the idea of the resampling method, we propose an algorithm for risk parity portfolio construction where the resampled portfolios are averaged with weights inversely proportional to the condition number of the Herfindahl index to mitigate the adverse impact of the covariance misspecification

to the out-of-sample risk contributions of constituents. Experimental results on factor datasets indicate that our method improves the risk parity strategy in terms of risk contributions of constituents. The traditional risk-adjusted return is also improved by our method in a modest but consistent manner, however without statistical significance.

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